

Stroke

Overview

A stroke is a medical emergency that occurs when the blood supply to a part of the brain is interrupted or significantly reduced, preventing brain tissue from receiving oxygen and nutrients. Brain cells begin to die within minutes of losing their blood supply. Immediate recognition and treatment are critical because delays can result in permanent brain damage, disability, or death.

Stroke is one of the leading causes of disability worldwide. Early diagnosis and timely medical intervention greatly improve recovery outcomes. Patients who experience sudden neurological symptoms should seek emergency medical care immediately.

Types of Stroke

There are three major types of stroke.

1. Ischemic Stroke

Ischemic stroke is the most common type, accounting for approximately 85% of all stroke cases. It occurs when an artery supplying blood to the brain becomes blocked by a blood clot or fatty deposits.

Common causes include:

- Atherosclerosis
 - Blood clots from the heart
 - Narrowed blood vessels
 - Carotid artery disease
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2. Hemorrhagic Stroke

A hemorrhagic stroke occurs when a weakened blood vessel ruptures, causing bleeding inside or around the brain.

Common causes include:

- High blood pressure
- Brain aneurysm

- Head injury
 - Blood vessel abnormalities
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3. Transient Ischemic Attack (TIA)

A transient ischemic attack, often called a "mini-stroke," occurs when blood flow to part of the brain is temporarily blocked.

Symptoms usually disappear within minutes or hours.

Although symptoms resolve, a TIA is a serious warning sign that a full stroke may occur in the future.

Risk Factors

Several medical conditions and lifestyle habits increase the risk of stroke.

Non-modifiable Risk Factors

- Increasing age
- Family history of stroke
- Male gender
- Previous stroke or TIA

Modifiable Risk Factors

- High blood pressure
- Diabetes mellitus
- High cholesterol
- Smoking
- Obesity
- Physical inactivity
- Excessive alcohol consumption
- Heart disease
- Atrial fibrillation

Managing these conditions can significantly reduce stroke risk.

Symptoms

Stroke symptoms usually appear suddenly.

Common symptoms include:

- Sudden weakness of one side of the body
- Sudden numbness of the face, arm, or leg
- Facial drooping
- Difficulty speaking
- Slurred speech
- Difficulty understanding speech
- Sudden confusion
- Loss of balance
- Difficulty walking
- Sudden dizziness
- Double vision
- Loss of vision in one or both eyes
- Severe headache without an obvious cause

Symptoms may affect either the left or right side of the body depending on the affected region of the brain.

FAST Recognition

The FAST method helps recognize stroke quickly.

F — Face Drooping

Ask the person to smile.

One side of the face may droop.

A — Arm Weakness

Ask the person to raise both arms.

One arm may drift downward or be difficult to lift.

S — Speech Difficulty

Ask the person to repeat a simple sentence.

Speech may be slurred, confused, or absent.

T — Time to Call Emergency Services

If any of these signs are present, emergency medical care should be sought immediately.

Rapid treatment improves the chances of recovery.

Diagnosis

Healthcare providers diagnose stroke using a combination of clinical examination and medical imaging.

Common diagnostic methods include:

- Neurological examination
- Blood pressure measurement
- Blood glucose testing
- CT scan of the brain
- MRI of the brain
- CT angiography
- Carotid ultrasound
- ECG
- Echocardiogram

- Blood tests

Imaging helps determine whether the stroke is ischemic or hemorrhagic.

Emergency Treatment

Treatment depends on the type of stroke.

Treatment for Ischemic Stroke

Possible treatments include:

- Intravenous thrombolytic medication (clot-busting medicine)
- Mechanical thrombectomy
- Antiplatelet medications
- Blood pressure control
- Oxygen therapy
- Monitoring in a stroke unit

Treatment is most effective when started within a few hours after symptom onset.

Treatment for Hemorrhagic Stroke

Treatment may include:

- Blood pressure management
 - Reversal of blood-thinning medications
 - Surgery to repair bleeding vessels
 - Removal of accumulated blood
 - Intensive care monitoring
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Rehabilitation

Recovery after stroke often requires long-term rehabilitation.

Rehabilitation may involve:

- Physical therapy
- Occupational therapy
- Speech therapy
- Swallowing assessment
- Cognitive rehabilitation
- Psychological counseling

Recovery varies depending on the severity of brain injury and how quickly treatment began.

Possible Complications

Stroke may result in long-term complications such as:

- Paralysis
- Muscle weakness
- Speech impairment
- Memory problems
- Difficulty swallowing
- Depression
- Chronic pain
- Balance problems
- Vision impairment

Early rehabilitation improves functional recovery.

Prevention

Many strokes can be prevented through healthy lifestyle choices.

Recommendations include:

- Control blood pressure
- Manage diabetes

- Stop smoking
 - Exercise regularly
 - Maintain a healthy weight
 - Eat a balanced diet
 - Reduce salt intake
 - Limit alcohol consumption
 - Take prescribed medications regularly
 - Attend routine medical checkups
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Patient Advice

Patients should seek immediate medical attention if they experience sudden neurological symptoms such as facial drooping, weakness of an arm or leg, slurred speech, severe headache, confusion, or vision loss.

Patients recovering from stroke should follow prescribed medications, participate actively in rehabilitation programs, attend follow-up appointments, and adopt healthy lifestyle habits to reduce the risk of future strokes.

Summary

Stroke is a serious neurological emergency caused by interruption of blood flow or bleeding within the brain. Early recognition of symptoms, rapid emergency treatment, and comprehensive rehabilitation significantly improve patient outcomes. Controlling risk factors such as hypertension, diabetes, smoking, and high cholesterol plays a major role in preventing stroke and reducing disability.